BẢNG TỔNG HỢP KẾT QUẢ BENCHMARK TOÀN DIỆN T-RAG v2

Báo Cáo Đánh Giá Chi Tiết 61 Cấu Hình Thử Nghiệm Trên Tập Dữ Liệu EnterpriseRAG-Bench (500 Câu Hỏi Multi-Hop/Single-Hop)

Môi Trường Thử Nghiệm & Tập Dữ Liệu Benchmark	Đóng Góp Kỹ Thuật Cốt Lõi Của T-RAG v2
• **Tập Dữ Liệu:** EnterpriseRAG-Bench (500 câu hỏi, 9 Shards, 4,213,106 Chunks). • **LLM Generator:** Qwen2.5-14B-Instruct (vLLM Engine, Temperature=0.0). • **Dense Retrieval:** LanceDB IVF-PQ (`BAAI/bge-large-en-v1.5`, 1024-dim). • **Sparse Retrieval:** Tantivy BM25 ($k_1 = 0.9, b = 0.4$, Trọng số Sparse $S = 0.7$). • **Cross-Encoder Reranker:** `BAAI/bge-reranker-large` (Top-K=5).	• **PSR Router:** Dự đoán xác suất Shard theo Entropy Shannon ($conf = 1 - H/\ln K$). • **Adaptive Tau:** Ngưỡng quét $\tau_{eff} = \tau_{base} + \alpha \cdot (conf - 0.5)$ tự động điều chỉnh. • **Smart Hop 2:** Bypass Hop 2 nếu $active\_shards < 2$ hoặc $dist < 0.55$ (giảm 34% latency). • **Zero-Copy Vector Reuse:** Tái sử dụng embedding Hop 1 cho Hop 2 (tăng tốc 6x truy xuất). • **CSEP:** Trích xuất thực thể ngữ cảnh theo Shard đã kích hoạt.

T-RAG v2 Configurations (47) **T-RAG v1 Pipelines (7)** **Standard Baselines (7)**

Corr%: Correctness | **Comp%**: Completeness | **Combined**: Corr × Comp | **Refused%**: Tỉ lệ từ chối

#	Pipeline / Cấu hình	Thông số Kỹ thuật Chi tiết	Corr	Comp	Combined	Refused	Total Lat	Retr Lat	Search Space
1	OPT: High Recall + Sparse Heavy	$\tau = 0.05, \gamma = 0.5, \alpha = 0.08, D = 0.3/S = 0.7$, CSEP, SmartHop2, AdaptTau	**36.80%**	45.67%	**32.24**	17.2%	1.02s	0.43s	3,577,666
2	Grid Tau=0.10	$\tau = 0.10, \gamma = 0.5, \alpha = 0.08, D = 0.5/S = 0.5$, CSEP, SmartHop2, AdaptTau	36.67%	44.99%	31.94	18.4%	0.88s	0.29s	3,323,143
3	OPT: Gamma=0.4	$\tau = 0.15, \gamma = 0.4, \alpha = 0.08, D = 0.5/S = 0.5$, CSEP, SmartHop2, AdaptTau	36.47%	45.23%	31.29	18.4%	0.83s	0.25s	2,706,911
4	Grid Gamma=0.0	$\tau = 0.15, \gamma = 0.0, \alpha = 0.08, D = 0.5/S = 0.5$, CSEP, SmartHop2, AdaptTau	36.40%	45.52%	31.86	17.0%	0.84s	0.25s	2,706,836
5	Grid Alpha=0.15	$\tau = 0.15, \gamma = 0.5, \alpha = 0.15, D = 0.5/S = 0.5$, CSEP, SmartHop2, AdaptTau	36.27%	45.47%	31.84	17.8%	0.86s	0.28s	3,006,387
6	Grid Alpha=0.50	$\tau = 0.15, \gamma = 0.5, \alpha = 0.50, D = 0.5/S = 0.5$, CSEP, SmartHop2, AdaptTau	36.07%	44.99%	31.40	17.4%	0.90s	0.33s	3,573,824
7	Grid Tau=0.05	$\tau = 0.05, \gamma = 0.5, \alpha = 0.08, D = 0.5/S = 0.5$, CSEP, SmartHop2, AdaptTau	36.07%	44.97%	31.24	17.4%	0.90s	0.33s	3,573,824
8	OPT: D=0.4/S=0.6	$\tau = 0.15, \gamma = 0.5, \alpha = 0.08, D = 0.4/S = 0.6$, CSEP, SmartHop2, AdaptTau	36.00%	**46.65%**	32.01	18.0%	0.89s	0.31s	2,706,057
9	Targeted D: Alpha Sweet Spot	$\tau = 0.05, \gamma = 0.4, \alpha = 0.15, D = 0.4/S = 0.6$, CSEP, SmartHop2, AdaptTau	36.00%	45.32%	31.10	17.2%	0.96s	0.36s	3,566,845
10	Grid Dense=0.3	$\tau = 0.15, \gamma = 0.5, \alpha = 0.08, D = 0.3/S = 0.7$, CSEP, SmartHop2, AdaptTau	35.80%	45.61%	31.85	18.6%	0.92s	0.36s	2,687,974
11	Grid Alpha=0.00	$\tau = 0.15, \gamma = 0.5, \alpha = 0.00, D = 0.5/S = 0.5$, CSEP, SmartHop2	35.67%	44.19%	30.69	18.0%	0.82s	0.25s	2,393,189
12	OPT: Best Completeness	$\tau = 0.15, \gamma = 0.5, \alpha = 0.04, D = 0.3/S = 0.7$, CSEP, SmartHop2, AdaptTau	35.60%	45.84%	31.38	19.8%	0.91s	0.33s	2,540,258
13	Targeted G: Minimalist Best	$\tau = 0.10, \gamma = 0.4, \alpha = 0.15, D = 0.3/S = 0.7$, CSEP, SmartHop2, AdaptTau	35.60%	45.87%	31.24	17.0%	0.99s	0.40s	3,558,410
14	Grid Tau=0.20	$\tau = 0.20, \gamma = 0.5, \alpha = 0.08, D = 0.5/S = 0.5$, CSEP, SmartHop2, AdaptTau	35.47%	44.50%	30.63	18.0%	0.80s	0.23s	2,219,777
15	Grid Alpha=0.12	$\tau = 0.15, \gamma = 0.5, \alpha = 0.12, D = 0.5/S = 0.5$, CSEP, SmartHop2, AdaptTau	35.27%	44.64%	31.00	17.8%	0.84s	0.27s	2,853,707
16	Ablation: No AdaptTau	$\tau = 0.15, \gamma = 0.5, \alpha = 0.00$ (forced), D=0.5/S=0.5, CSEP, SmartHop2	35.27%	44.27%	30.48	18.0%	0.82s	0.24s	2,393,189
17	OPT: Low Latency	$\tau = 0.20, \gamma = 0.3, \alpha = 0.00, D = 0.5/S = 0.5$, CSEP, SmartHop2, AdaptTau	35.27%	43.71%	30.21	19.0%	0.76s	0.20s	1,964,143
18	OPT: Speed+Sparse	$\tau = 0.20, \gamma = 0.5, \alpha = 0.00, D = 0.3/S = 0.7$, CSEP, SmartHop2, AdaptTau	35.20%	44.65%	30.68	18.4%	0.85s	0.27s	1,944,632
19	OPT: Balanced	$\tau = 0.15, \gamma = 0.5, \alpha = 0.04, D = 0.5/S = 0.5$, CSEP, SmartHop2, AdaptTau	35.07%	45.21%	30.89	18.6%	0.83s	0.26s	2,562,116
20	Grid Alpha=0.04	$\tau = 0.15, \gamma = 0.5, \alpha = 0.04, D = 0.5/S = 0.5$, CSEP, SmartHop2, AdaptTau	35.07%	45.20%	30.77	18.6%	0.83s	0.26s	2,562,116
21	Grid Alpha=0.25	$\tau = 0.15, \gamma = 0.5, \alpha = 0.25, D = 0.5/S = 0.5$, CSEP, SmartHop2, AdaptTau	34.87%	44.37%	30.62	17.2%	0.90s	0.32s	3,416,644
22	Grid Gamma=1.0	$\tau = 0.15, \gamma = 1.0, \alpha = 0.08, D = 0.5/S = 0.5$, CSEP, SmartHop2, AdaptTau	34.87%	42.32%	29.71	19.0%	0.83s	0.27s	2,704,042
23	OPT: D=0.2/S=0.8	$\tau = 0.15, \gamma = 0.5, \alpha = 0.08, D = 0.2/S = 0.8$, CSEP, SmartHop2, AdaptTau	34.80%	45.08%	30.83	18.8%	0.97s	0.39s	2,693,001
24	T-RAG v2 Standard	$\tau = 0.15, \gamma = 0.5, \alpha = 0.08, D = 0.5/S = 0.5$, CSEP, SmartHop2, AdaptTau	34.67%	44.91%	30.72	17.0%	0.84s	0.27s	2,711,818
25	Ablation: No CSEP	$\tau = 0.15, \gamma = 0.5, \alpha = 0.08, D = 0.5/S = 0.5$, No CSEP, SmartHop2, AdaptTau	34.67%	44.38%	30.13	18.2%	0.82s	0.25s	2,704,197
26	Grid Dense=0.1	$\tau = 0.15, \gamma = 0.5, \alpha = 0.08, D = 0.1/S = 0.9$, CSEP, SmartHop2, AdaptTau	34.60%	45.13%	30.67	17.8%	1.02s	0.42s	2,690,576
27	Targeted H: Speed King v2	$\tau = 0.15, \gamma = 0.0, \alpha = 0.00, D = 0.4/S = 0.6$, CSEP, SmartHop2	34.60%	43.75%	30.32	19.0%	0.84s	0.26s	2,387,237
28	OPT: Retrieve Depth=30	$\tau = 0.15, \gamma = 0.5, \alpha = 0.08, R = 30, K = 5$, CSEP, SmartHop2, AdaptTau	34.47%	44.42%	30.38	**16.8%**	0.92s	0.34s	2,707,042
29	Grid Gamma=0.3	$\tau = 0.15, \gamma = 0.3, \alpha = 0.08, D = 0.5/S = 0.5$, CSEP, SmartHop2, AdaptTau	34.47%	44.54%	30.19	18.0%	0.82s	0.25s	2,706,695

#	Pipeline / Cấu hình	Thông số Kỹ thuật Chi tiết	Corr	Comp	Combined	Refused	Total Lat	Retr Lat	Search Space
30	Grid Gamma=0.7	$\tau = 0.15, \gamma = 0.7, \alpha = 0.08, D = 0.5/S = 0.5$, CSEP, SmartHop2, AdaptTau	34.07%	44.39%	29.84	17.6%	0.83s	0.25s	2,698,037
31	OPT: Best Correctness	$\tau = 0.15, \gamma = 0.5, \alpha = 0.25, D = 0.1/S = 0.9$, CSEP, SmartHop2, AdaptTau	34.00%	44.03%	29.67	19.2%	1.07s	0.47s	3,392,562
32	T-RAG v1 High-Recall G1	$\tau = 0.05, \gamma = 1.0$, No CSEP, No AdaptTau	33.80%	43.43%	29.71	22.6%	1.19s	2.31s	3,550,005
33	T-RAG v1 Balanced G1	$\tau = 0.15, \gamma = 1.0$, No CSEP, No AdaptTau	33.80%	43.29%	29.66	20.6%	1.02s	1.48s	2,354,016
34	OPT: High Speed Sparse Heavy	$\tau = 0.30, \gamma = 0.5, \alpha = 0.08, D = 0.1/S = 0.9$, CSEP, SmartHop2, AdaptTau	33.60%	44.77%	29.52	19.8%	0.86s	0.29s	1,438,218
35	Hybrid Baseline	BM25 + Vector Hybrid, Reranker, K=5	33.60%	44.11%	29.35	18.0%	1.15s	0.63s	4,213,106
36	Grid Tau=0.30	$\tau = 0.30, \gamma = 0.5, \alpha = 0.08, D = 0.5/S = 0.5$, CSEP, SmartHop2, AdaptTau	33.27%	42.91%	29.33	18.0%	0.74s	0.18s	1,456,839
37	Sparse Only	$\tau = 0.15, \gamma = 0.5, \alpha = 0.08, D = 0.0/S = 1.0$, CSEP, SmartHop2, AdaptTau	33.20%	43.77%	29.88	20.0%	1.07s	0.48s	2,691,477
38	OPT: Max Performance	$\tau = 0.10, \gamma = 0.5, \alpha = 0.04, D = 0.1/S = 0.9$, CSEP, SmartHop2, AdaptTau	33.20%	43.67%	29.17	18.8%	1.04s	0.45s	3,100,597
39	Targeted B: Precision Strike	$\tau = 0.10, \gamma = 0.0, \alpha = 0.15, D = 0.4/S = 0.6$, CSEP, SmartHop2, AdaptTau	33.20%	43.72%	28.59	19.2%	0.96s	0.36s	3,566,651
40	T-RAG v1 High-Speed G1	$\tau = 0.30, \gamma = 1.0$, No CSEP, No AdaptTau	33.00%	41.86%	29.15	21.4%	0.93s	1.08s	**1,282,848**
41	HyDE Baseline	HyDE (LLM Hypothetical Gen) → Hybrid → Rerank	33.00%	42.97%	28.01	18.2%	1.23s	0.63s	4,213,106
42	Ablation: No Smart Hop2	$\tau = 0.15, \gamma = 0.5, \alpha = 0.08, D = 0.5/S = 0.5$, Always Hop2 ON	32.80%	43.93%	29.39	18.6%	1.16s	0.58s	2,681,805
43	Query Expansion Baseline	LLM 3 Sub-queries → Parallel Hybrid → RRF → Rerank	32.60%	43.14%	28.34	19.8%	2.46s	1.91s	4,213,106
44	Targeted A: Ultimate Combo	$\tau = 0.05, \gamma = 0.0, \alpha = 0.15, D = 0.3/S = 0.7$, CSEP, SmartHop2, AdaptTau	32.40%	42.77%	28.41	21.4%	1.00s	0.39s	3,591,459
45	T-RAG v1 High-Recall	$\tau = 0.05, \gamma = 0.0$, No CSEP, No AdaptTau	32.40%	43.25%	28.15	19.6%	1.25s	2.28s	3,598,130
46	T-RAG v1 Balanced	$\tau = 0.15, \gamma = 0.0$, No CSEP, No AdaptTau	32.40%	42.56%	27.91	19.2%	1.05s	1.53s	2,372,449
47	OPT: Retrieve Depth=10	$\tau = 0.15, \gamma = 0.5, \alpha = 0.08, R = 10, K = 5$, CSEP, SmartHop2	32.40%	42.29%	27.85	22.2%	0.71s	**0.16s**	2,711,610
48	T-RAG v1 High-Speed	$\tau = 0.30, \gamma = 0.0$, No CSEP, No AdaptTau	32.20%	42.44%	28.36	22.0%	0.92s	1.01s	1,284,666
49	Targeted F: Wide Net Balanced	$\tau = 0.05, \gamma = 0.0, \alpha = 0.08, R = 25, K = 5$, CSEP, SmartHop2	32.20%	43.34%	27.79	19.4%	1.01s	0.42s	3,575,488
50	Targeted E: Low Gamma High Alpha	$\tau = 0.10, \gamma = 0.0, \alpha = 0.12, D = 0.3/S = 0.7$, CSEP, SmartHop2	31.80%	43.46%	28.17	21.2%	1.00s	0.39s	3,483,602
51	T-RAG v1 No Reranker	$\tau = 0.15, \gamma = 0.0$, No Reranker, No CSEP, No AdaptTau	31.80%	40.72%	27.69	20.8%	1.06s	1.66s	2,372,449
52	Targeted C: Gamma Zero Low Tau	$\tau = 0.10, \gamma = 0.0, \alpha = 0.08, D = 0.3/S = 0.7$, CSEP, SmartHop2	31.00%	43.51%	27.55	21.2%	1.00s	0.38s	3,323,775
53	OPT: TopK=3	$\tau = 0.15, \gamma = 0.5, \alpha = 0.08, K = 3$, CSEP, SmartHop2	30.60%	42.06%	27.39	21.6%	0.66s	0.26s	2,711,818
54	BM25 Baseline	Pure BM25, K=5	29.20%	39.50%	25.23	23.0%	0.97s	0.41s	4,213,106
55	Grid Dense=0.7	$\tau = 0.15, \gamma = 0.5, \alpha = 0.08, D = 0.7/S = 0.3$, CSEP, SmartHop2	28.26%	37.18%	24.34	24.0%	0.82s	0.26s	2,704,805
56	LLM Router Baseline	LLM Prompting Router → Hybrid → RRF → Rerank	27.80%	38.57%	23.96	21.8%	0.88s	0.32s	2,075,036
57	Grid Dense=0.9	$\tau = 0.15, \gamma = 0.5, \alpha = 0.08, D = 0.9/S = 0.1$, CSEP, SmartHop2	25.85%	34.83%	22.14	25.4%	0.82s	0.25s	2,704,805
58	VECTOR_RERANKER Baseline	Pure Vector → Cross-Encoder Reranker, K=5	24.65%	33.84%	20.89	26.8%	0.69s	0.23s	4,213,106
59	Dense Only	$\tau = 0.15, \gamma = 0.5, \alpha = 0.08, D = 1.0/S = 0.0$, CSEP, SmartHop2	24.25%	34.66%	21.11	26.2%	0.82s	0.25s	2,705,415
60	OPT: TopK=1	$\tau = 0.15, \gamma = 0.5, \alpha = 0.08, K = 1$, CSEP, SmartHop2	22.40%	33.05%	19.63	40.6%	**0.48s**	0.25s	2,711,818
61	Vector Baseline	Pure Vector, K=5	19.60%	30.24%	16.47	32.0%	0.60s	0.20s	4,213,106